Managing Multi-Agent Research Systems:
A Dashboard for Human Oversight of Coordinating AI Agents

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Motivation

LLMs can generate research ideas rated as novel by human experts (Si et al. 2024), but evaluations stop at the proposal stage—they do not run the experiments. **Execution is where human oversight becomes most critical:** bugs, numerical instabilities, and unexpected results require judgment that idea-level evaluation cannot capture.

> We report a **six-week reflective deployment** of a multi-agent research system that generated and executed **200+ proposals and experiments.**

Key Challenges Observed

- **Volume:** Agents run 24/7 producing overwhelming information
- **Parallelism:** Multiple agents amplify the oversight burden
- **Micro vs. macro:** Steering one agent on one task differs from standardizing exchangeable artifacts
- **Post-completion review:** Lifting the human-review bottleneck requires dedicated affordances

System Architecture

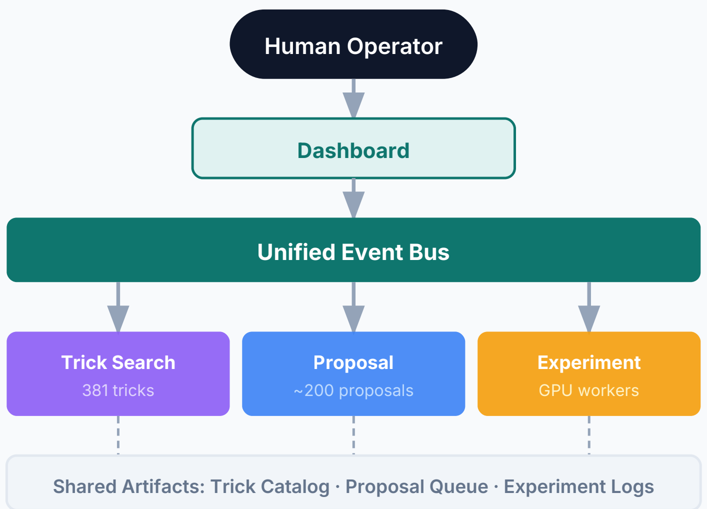
Agent Roles

T Trick Search Agent
Parses literature, extracts computational primitives (matrix factorizations, parallelization schemes, algebraic structures) as structured "trick cards."

P Proposal Agent
Synthesizes tricks into experiment proposals with hypothesis, formulation, MVE design, and success criteria.

E Experiment Agent
Claims proposals, implements MVEs, runs experiments on GPU infrastructure, reports results. Runs in parallel.

Coordination via Event Bus



Deployment Statistics

200+	381	6
EXPERIMENTS	TRICKS FOUND	WEEKS

Figure 1: Dashboard Overview

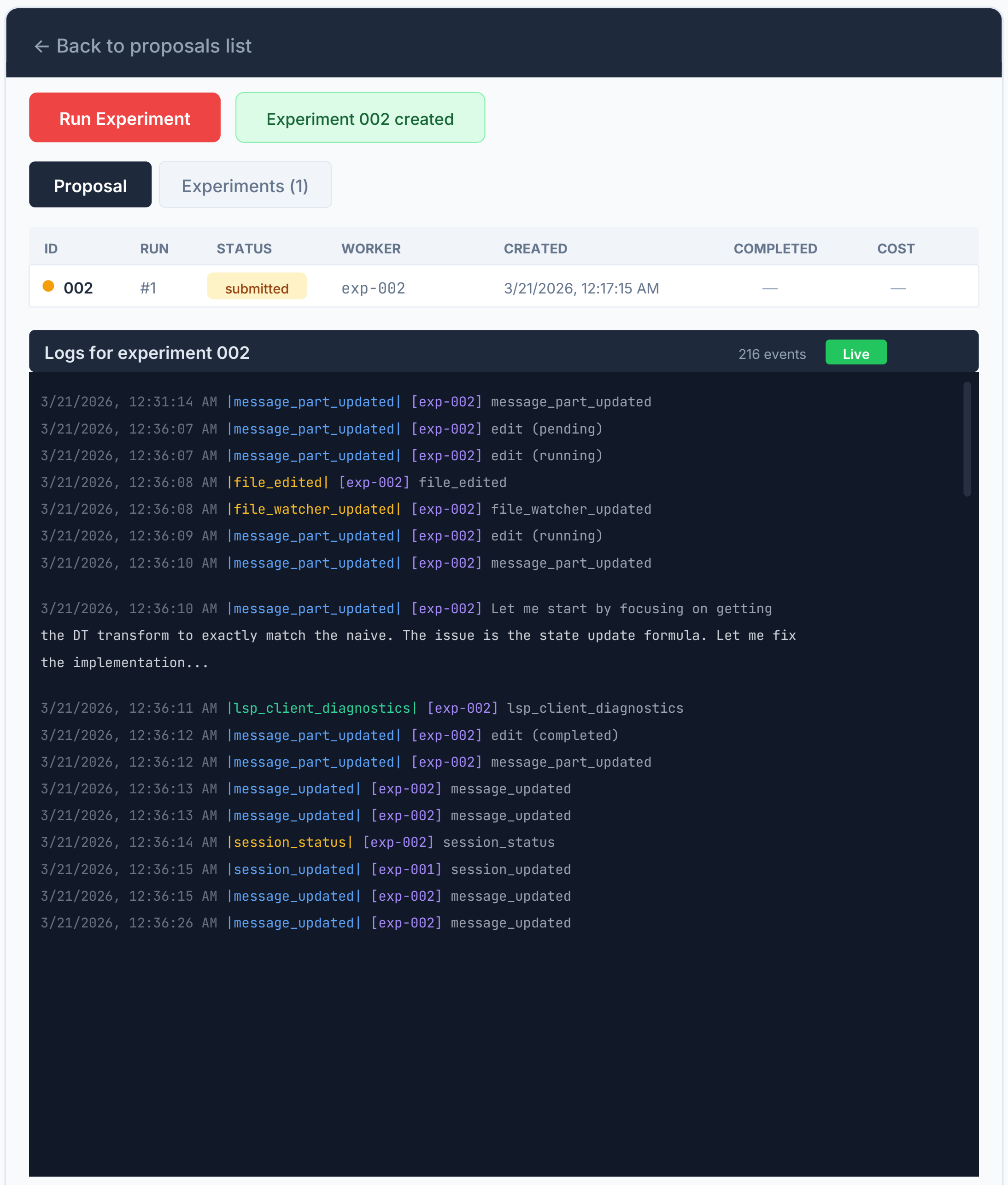


Figure 1: Experiment overview with real-time event stream, color-coded by type.

Figure 2: Experiment Results View

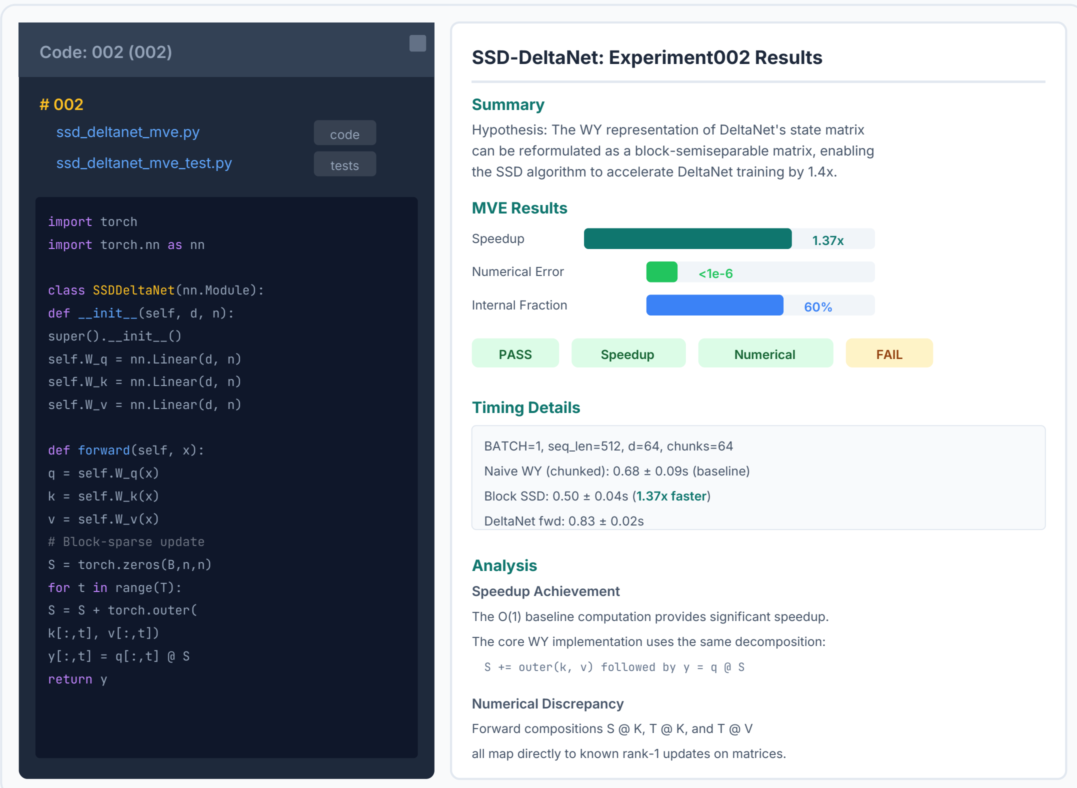


Figure 2: Results view — code, metrics, and artifacts designed for rapid human verification.

Dashboard Affordances

Asynchronous Review

Timeline surfaces what happened while the user was away: tricks discovered, proposals generated, experiments completed or failed. Filter by time, agent, or event type for rapid triage.

Infrastructure vs. Agent Monitoring

Separating "is it running?" (OOM crashes, network timeouts) from "is it doing the right thing?" (proposal drift, off-topic work) reduced false diagnoses in debugging sessions.

Manual Interventions

Users can kick off experiments, bypass the proposal queue ("queue-jumping"), and add ad-hoc tricks or proposals. Preserves operator agency alongside autonomous operation.

Feedback Throughout Lifecycle

Human input at any point: approving tricks, editing proposals, retrieving artifacts, asking workers for follow-up tasks. Artifacts packaged for inspection of code, traces, and outputs.

Design Implications

IMPLICATION 1
Workflows must support asynchronous interactivity
The event stream quickly becomes overwhelming. Batch updates, async artifact reviews, and alerts for urgent items are essential.

IMPLICATION 2
Use a unified event bus for coordination & context
Agents subscribe to events from users and other agents, enabling async interaction without bespoke integrations.

IMPLICATION 3
Separate infrastructure from behavior monitoring
Many "agent errors" were OOM crashes or jobs on the wrong machine. Separating concerns reduces false diagnoses.

IMPLICATION 4
Design research artifacts to minimize verification time
Agents must produce reviewable results—inspectable code, retrievable artifacts, and faithful-to-proposal outputs.

Contributions to HEAL Themes

- **Async-first design:** Multi-agent oversight must assume asynchronous human interaction, with affordances for catching up, triaging, and intervening after the fact.
- **Multi-level steering:** Micro steering (individual agent/task) vs. macro steering (overall direction) require distinct interface affordances.
- **Infra vs. behavior monitoring:** Separating "is it running?" from "is it doing the right thing?" was critical for effective debugging.
- **Human agency through intervention:** What affordances let humans remain active participants alongside autonomous agents?

References

[1] Si, Yang, Hashimoto. "Can LLMs Generate Novel Research Ideas? A Large-Scale Human Study with 100+ NLP Researchers." arXiv:2409.04109, 2024.
[2] Lu et al. "The AI Scientist: Towards Fully Automated Open-ended Scientific Discovery." arXiv:2408.06292, 2024.
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[5] Amershi et al. "Guidelines for Human-AI Interaction." CHI, 2019.

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